Actual cofactor equations force a full nonzero norm

2026-11-13. Unsigned producer evidence [pcc:unverified].

The actual reduced coefficient cofactor equations now imply

$$(C\alpha\, F)^2 - f(C\beta\, H)^2 = Cc\, G, \qquad c \neq 0.$$

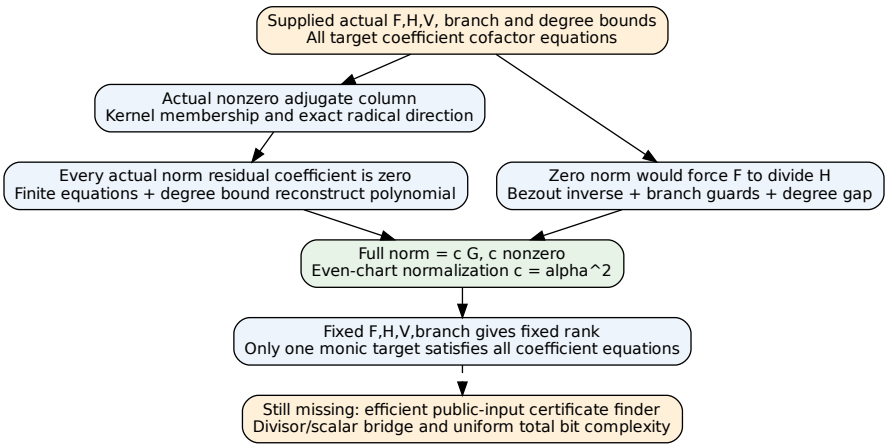
No norm identity, nonzero-normalization oracle, numerical rank or coordinate oracle is supplied to the final theorem. The prior actual branch construction derives its matrix guards. Suitable F, H, V and the cofactor equations are still inputs. Frozen source `5e7f9bb5c`; `NormCertificateRecovery`.

Exact hypotheses. Over a field: F, H, G monic, $\deg H > 0$, $\deg H + 1 < \deg F$, $UF + V(fH) = 1$, $F(q) \neq 0$, $\alpha \neq 0$, $\alpha F(q) - y\beta H(q) = 0$, and

$$2\deg F \leq \deg G, \qquad \deg f + 2\deg H \leq \deg G.$$

For each $j < \deg G$, pair the actual reduced residual coefficient matrix $Q_{\mathrm{red},j}$ with the adjugate of the actual quotient-moment matrix S_{red}. All these pairings must be zero. The norm identity itself is not among the assumptions [pcc:unverified].

Noncircular extraction. A nonzero diagonal cofactor gives an actual nonzero adjugate column. The matrix-adjugate identity places it in the actual kernel. The exact radical line makes it a nonzero scalar multiple of the reduced candidate polynomial direction. Rank-one adjugate factorization and nonzero scaling transfer all pairings to the actual quadratic residuals. Public branch insertion gives every residual coefficient of the original norm. The degree bounds and coefficient extensionality reconstruct the full polynomial [pcc:unverified].



Nonzero normalization. The branch guards imply $\beta \neq 0$. A zero norm would give fH^2 as a constant multiple of F^2. Multiply the Bezout identity by H: then $F \mid H$, contradicting the degree gap. Hence the reconstructed scalar is nonzero. It is a polynomial normalization, not the secret discrete logarithm [pcc:unverified].

Exact normalization, falsification control and remaining finder

Even chart. If $\deg F = \deg H + 2$, $\deg f \leq 3$ and $\deg G = 2\deg F$, the lower norm term has degree strictly below $2\deg F$. Monicity makes the top coefficient exactly α^2. Thus the actual normalization is $c = \alpha^2 \neq 0$. The old normOfWitnessLine theorem, which already assumes a norm identity, is not used circularly [pcc:unverified].

Target control. Fixed F, H, V, q, y give a fixed reduced moment matrix; it does not depend on G. A fixed nonzero norm has at most one monic target. Combining this uniqueness lemma with actualNonzeroNorm, at most one monic G can satisfy all actual cofactor equations under the stated bounds. Rank alone is therefore an insufficient experimental gate [pcc:unverified].

Native regression. On the existing F17 example, both fixed supports retain the same curve, branch $q = 0, y = 6$, monic degree 6 and distinct rational nonzero-y lifts. The same reduced matrix has rank 4 and determinant 0. The changed set replaces 13 by 7; it is wrong for this certificate, not asserted globally unsatisfiable. One artifact invocation contains two controls [pcc:unverified].

Support	Rank	Cofactor pairings modulo 17
0,3,5,6,10,13	4	0,0,0,0,0,0
0,3,5,6,7,10	4	0,12,4,12,16,5

Pairings equal 2 times the actual norm residuals. Checked literal congruence, Gaussian rank versus independent Leibniz determinant/adjugate, both adjugate products, rational lifts and all chart guards. Three tests, strict clippy and formatting pass. Native source `582e21bfb`, fixed-case contract `ad3005105`; unsigned data `PCC-20261113-1aa548`. This is a finite development regression, not timing, relation-yield or independent A/A evidence.

Exact component	Declaration in `NormCertificateRecovery`
Actual residual coefficients	`actualNormResidual`
Full nonzero norm	`actualNonzeroNorm`
Even-chart normalization	`actualNormalizedNorm`
Monic target uniqueness	`monicScalarTargetUnique`

Verification scope. There are 16 new statements. Builds 001–005 all pass with explicit exits; no new-module warning. Whole-study tracked-source audit: 217 statements, 0 additional trusted axioms. The import-inclusive fresh replay has a separate completion receipt. Exact source bindings, logs and input hashes accompany this report. Producer checks do not supply independent signed PCC standing. No solver, timing, paid experiment, benchmark, leaderboard or ecdb row is added. Native clippy was initially absent; installed the component. The globally ignored lockfile was tracked before execution. No proof or native arithmetic check failed.

Remaining obligations. The polynomial certificate is now connected to its full nonzero norm for this chart. Prime subgroup, rational factor-base support, divisor/group relation and discrete-log recovery are additional application obligations. Efficiently finding the supplied F, H, V and target cofactor equations remains the algorithmic gap. A sub-exponential claim needs uniform total expected $\exp(o(\log r))$ work, including setup, rejected attempts, relation yield, linear algebra and final recovery. A certificate checker is not that finder. Reachability through the annotated jet completion remains separate and unproved; its H is not the norm polynomial by naming.

Sources. Prior branch, cofactor, extraction, norm and coefficient sources; pinned polynomial monic/degree/divisibility APIs; existing F17 control. KB unavailable. Standard algebra, no priority or algorithmic novelty claim. Exact paths, source bindings and reproduction commands are in the README.